

Only option D is possible.

- 6 D** $p_2 > p_1$
Upthrust is the product of the pressure difference and the cross-sectional area.
$$U = p_2 A - p_1 A = (p_2 - p_1) A = Ah\rho_L g$$

where ρ_L = density of the liquid (which is not provided in the question!)

- 7 A** Since the system is in equilibrium, the mass of the non-uniform rule can be found as